DEEP LEARNING CA3 · PRESENTATION SCRIPT

CryptoGraph Presentation Script

What to say, in what order, and what to click on the deployed dashboard: a 18-minute talk with an 8-minute cut, then the questions you are most likely to get, with short spoken answers. Plain words throughout; the professor's technical terms are in **bold** so you recognise them.

18 min	8 min	5 tabs	54
full talk, 15 segments	short version	deployed dashboard demo	likely questions, with answers

START HERE

Before you walk in

How to use this script, a checklist for the room, who says what, and the timetable.

Read the whole script out loud once with a timer, then a second time with the dashboard open and clicking along. Three kinds of boxes:

> **SAY**
>
> Lines to say. Learn the meaning, not the exact words; your own words sound more confident.

> **IF ASKED**
>
> One- or two-line answers to follow-up questions that usually come at that point.

ON SCREEN cues tell you what to show. Part numbers in brackets, such as (guide Part 7), point to the big study guide, `CryptoGraph_Viva_Prep_Guide.html`, for depth.

Checklist

1. **Open the deployed Streamlit link 10 minutes early** and click every tab once. Free Streamlit apps can fall asleep and take a minute to wake. Set *Look ahead* to 6 hours.

2. Use the **deployed link**, not the copy on your laptop: the laptop copy has extra, unreleased tabs from later work.
3. Keep a folder of **backup screenshots**, one per tab, in case the internet fails.
4. Open the report PDF at Table III (page 5) and Figure 2 (page 4), in case the professor wants to look at them.
5. Agree who says what (below) and who drives the mouse. Agree the answer to "who did what" so both of you say the same thing.
6. Phones silent, water nearby. Speak slower than feels natural.

Who says what (a suggestion)

Split by who built what: you (Hardik) built the data pipeline, the network analysis, the models, the evaluation and the explanations; Dakshesh built the prediction command, the dashboard and its crisis explorer, the model compression and the extra attack tests. Swap freely; what matters is that your answers match.

Run of show (18:00; for a 15-minute slot, move segments 6, 8 and 12 into questions and keep the demo to 3 minutes)

#	Segment	Time	Clock	Who	On screen
1	Opening	0:35	0:35	Hardik	dashboard, top of page
2	The problem	0:55	1:30	Hardik	dashboard
3	The data	0:50	2:20	Hardik	dashboard
4	What the network showed	1:30	3:50	Hardik	Market map tab
5	How we turned prices into events	1:45	5:35	Hardik	"What just happened"
6	How we judge a model	0:45	6:20	Hardik	–
7	The model: TGN	1:40	8:00	Hardik	report Figure 2
8	Why this design; the other models	0:50	8:50	Hardik	–
9	Results	1:15	10:05	Hardik	report Table III
10	After the report	1:30	11:35	Hardik	the before/after table
11	What drives an alert; how long shocks last	0:55	12:30	Dakshesh	–
12	The "Secure" part	0:35	13:05	Dakshesh	–
13	**Dashboard demo**	**3:45**	**16:50**	**Dakshesh**	**all 5 tabs**
14	Limits and next steps	0:45	17:35	Dakshesh	–
15	Close	0:25	18:00	Hardik	dashboard

THE SHORT VERSION (ABOUT 9½ MINUTES)

Keep segments 1, 2, 4 (network, 1 minute), 5 (events, 1 minute), 7 (TGN, 1:30), 9 (results, 1 minute), 10 (after the report, 1 minute), the demo (2:30: opening screen, Why, the bitcoin ETF crisis) and 15: about 9:30 in all. Drop the rest; their content is in the answers at the end, ready for questions.

SEGMENT 1 · 0:35 · HARDIK

Opening: the hook

Grab attention with real crashes, then state the question in one sentence.

ON SCREEN the deployed dashboard, top of the page.

> **SAY**
>
> "Professor, this project started from your suggestion: look at crypto contagion through networks. Crypto markets never close, and they react to news within hours. When Terra's LUNA collapsed in May 2022, when the FTX exchange failed in November 2022, and on 10 October 2025, most big coins fell within hours of each other.
>
> Most research, including the network measures we started from, describes how connected coins are. We took your idea one step further and asked: **when one coin makes a sudden big move, which other coins will be hit next, and how soon?** Our project, CryptoGraph-Secure, answers it with a **Temporal Graph Network** and shows the answer on this dashboard."

SEGMENT 2 · 0:55 · HARDIK

The problem statement

The exact task, and why it is hard.

SAY

"Precisely: when coin *i* has a **shock** at hour *t*, then for every other coin *j* we predict whether *j* also has a shock within the next 1, 3, 6, 12 or 24 hours. That is a yes-or-no question for each pair of coins and each time window.

Three things make it hard. First, it is **rare**: at one hour only about 10% of the answers are yes, so plain accuracy is useless. Second, everything depends on **time order**, so the model must never see the future. Third, most shocks are driven by **news**, which no model can see coming. So our goal is not perfect prediction. It is to rank the coins at risk clearly better than chance, and to measure that honestly."

IF ASKED

Who would use this? An exchange's risk team, a regulator, or a trader who wants a warning before trouble spreads.

Why "Secure"? Because we also test how the model holds up when inputs are missing, noisy or deliberately manipulated, and whether an anomaly detector can flag unusual market days.

SEGMENT 3 · 0:50 · HARDIK

The data

Where it comes from, how it was checked, and how time was split.

SAY

"We used hourly prices from **Binance's public archive**: 30 major coins from January 2020 to September 2026, 59,088 hours. All 3,049 files were checked against their SHA-256 fingerprints. The coins are the largest by market size with enough history on Binance; stablecoins are left out.

We handled real-world traps: MATIC renamed to POL, the LUNA name reused for a new coin after the collapse, delistings, and exchange maintenance gaps, which are masked rather than invented.

Time is split strictly: **training** up to June 2022, **validation** from July 2022 to December 2023, and a final **test** from January 2024 to September 2026, with a one-week gap, an **embargo**, at each border. Everything we fitted used the training years only."

IF ASKED

Why hourly, not daily? Contagion happens within hours: right after a shock, another coin is 2.3 times more likely to shock in the very next hour than in a random hour. Daily data would merge several shocks and their order into one number.

Why fixed dates, not a random split? A random split would let the model learn from the future of the same crash. Fixed dates also make sure each period contains a big crash: COVID and LUNA in training, FTX in validation, October 2025 in test.

Why not before 2020? Binance's archive goes back to August 2017, but only 15 of our 30 coins were listed before 2020, so the network would be much smaller in those years.

SEGMENT 4 · 1:30 · HARDIK

What the network analysis showed

The professor's own idea, so present it as findings, including the ones that surprised you.

ON SCREEN the dashboard's Market map tab, or notebook 02's MST and FTX charts.

SAY

"Your suggestion to look at contagion through networks shaped the whole project, so here is what the networks told us. We built three: who **moves together** (correlation over the past week), who **passes uncertainty** to whom (the Diebold–Yilmaz spillover network), and who **shocks after** whom (a shock-transmission network built from time order).

Four findings. One: a single market factor explains 64% of hourly movement; the biggest eigenvalue is 19.3 out of 30. Two: in all nine crises the network tightens: correlations rise, the minimum spanning tree shrinks, and spreading money across coins stops protecting you. Three: groups of coins partly match their sectors, but contagion ignores sectors, except meme coins.

Four, and this one surprised us: at hourly speed there is almost no leader. Bitcoin does not move first; the correlation one hour later is tiny, 0.035, and the hourly spillover ranking turned out to be the same as plain centrality. In the FTX collapse, 18 to 24 coins shocked in the *same* hour as FTT's biggest falls. So contagion here is mostly simultaneous and news-driven, not a chain from coin to coin.

That is why we did not draw fixed 'who infects whom' arrows. We used the network as features that change over time, let the TGN learn from the event stream, and turned the shock-transmission network into our baseline, an independent cascade."

CAREFUL

Present "no hourly leader" as a *finding we tested four ways*, not as "the network idea didn't work". The network is still in the model (8 features per coin), it gave the baseline, and it explains why contagion looks the way it does.

SEGMENT 5 · 1:45 · HARDIK

How we turned prices into events

The answer to "how did you quantify events?", in four steps. The question comes in almost every viva.

ON SCREEN "What just happened" at the top of the dashboard: it shows exactly this, a shock and its size in "times its usual move".

> **SAY**
>
> "**Step one, define a shock.** For every coin and hour we take the hourly return, remove the normal rush-hour pattern of the market, and divide by the coin's typical hourly move over the previous week: 168 hours, ending the hour before. That gives a score called **z**, how many normal moves big this move is. If z is bigger than 2.29, up or down, it is a shock. We chose 2.29 on training data so that about 4% of hours are shocks.
>
> **Step two, every shock becomes events.** If SOL shocks at 2 pm, we create one event for every other coin trading then: SOL to BTC, SOL to ETH, and so on, about 29 events.
>
> **Step three, describe each event.** Each event carries 5 numbers: how closely the two coins moved over the past week, in the same hour and one hour apart; the shocking coin's return; its volatility; and a shock flag. Each coin also has 14 features: 6 about its recent trading, such as volatility and volume compared with its own last week, and 8 about its position in the network of coins.
>
> **Step four, write down the answers.** For each event, five yes-or-no labels: did coin *j* have its own shock within 1, 3, 6, 12 and 24 hours? Strictly later, never in the same hour, because moving in the same hour is co-movement, not contagion.
>
> That gives **1.46 million events** from 2020 to 2026."

A worked example (made-up numbers, for explaining only)

Event at 14:00 UTC	What happened next	1 h	3 h	6 h	12 h	24 h
SOL crashes, z = −3.4	(the trigger, not an event itself)					
SOL → ETH	ETH shocks at 15:00	1	1	1	1	1
SOL → BTC	BTC shocks at 19:00	0	0	1	1	1
SOL → DOGE	DOGE shocks only at 14:00, same hour	0	0	0	0	0
… 26 more coins	one event each	…	…	…	…	…

IF ASKED

Why divide by the past week's volatility? So each move is judged against that coin's own normal. A 3% hour is huge for Bitcoin and ordinary for a meme coin. It also follows Forbes and Rigobon's warning that contagion must be measured against normal co-movement.

Why up to the hour before? So a move is never judged partly against itself; our first draft made exactly that mistake and we fixed it.

Why 168 hours? One full week covers every weekday and weekend: long enough to be stable, short enough to adapt.

How often is the answer yes? On test: 10.5%, 20.6%, 31.0%, 44.7% and 62.7% for 1, 3, 6, 12 and 24 hours.

The shock flag is always 1? Yes: every event starts with a shock, so it carries no information; it marks the event type. The four informative numbers are the two correlations, the return and the volatility.

SEGMENT 6 · 0:45 · HARDIK

How we judge a model

Why AUPRC, and the rules that keep the comparison fair.

SAY

"Because only 10% of answers are yes, a model that always says no is 90% accurate and completely useless. So we use **AUPRC**, the area under the precision–recall curve. It rewards putting the real yes-cases at the top of the list. Random guessing scores the share of yes-cases, 0.105 at one hour, so every number is compared with that.

Fairness rules: every model is scored on exactly the same events; we tuned only on validation; we scored the test period once; we trained five seeds; and we tested differences with a **bootstrap** that resamples whole days, because one crash creates hundreds of linked events."

IF ASKED

Precision and recall? Precision: of our alarms, how many were right. Recall: of the real events, how many we caught. AUPRC sums up the trade-off at every threshold.

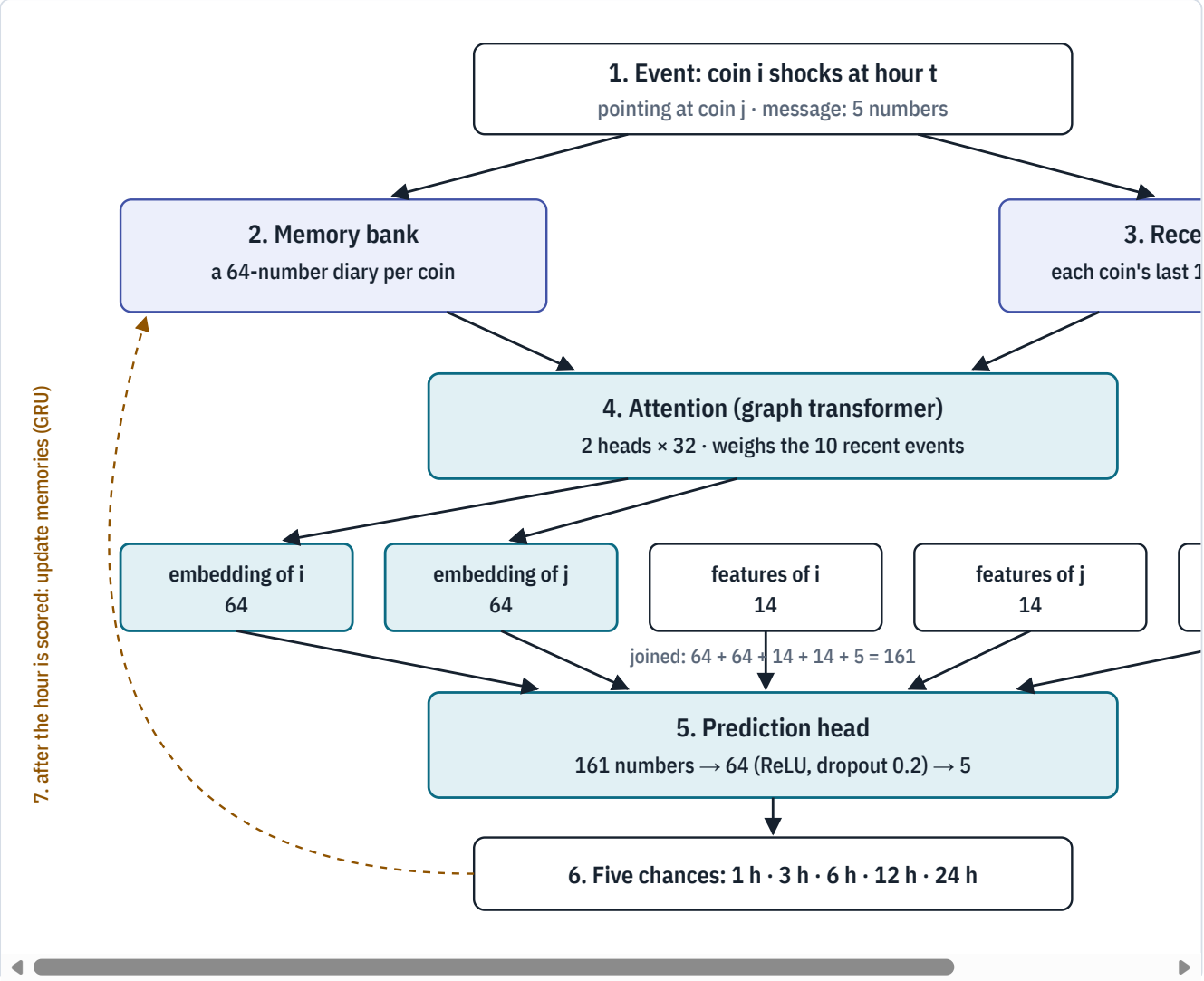
Why not AUROC? We report it too (0.64 to 0.69), but with rare positives AUROC can look good while most alarms are false; precision–recall is more informative (Davis and Goadrich, 2006).

SEGMENT 7 · 1:40 · HARDIK

The model: the Temporal Graph Network

What it is, who made it, how it works box by box, and its size.

ON SCREEN report Figure 2 (page 4), or the diagram below.



How the TGN scores one event, in seven steps. The dashed amber loop is the only way the future reaches the memory: an hour's events are written in only after the whole hour has been predicted.

SAY

"Our main model is the **Temporal Graph Network**, published by Rossi and colleagues at Twitter in 2020 for streams of time-stamped interactions. Think of it as a diary for every coin.

One: every coin has a **memory** of 64 numbers, its diary. Two: when an event arrives, coin *i* shocked and we ask about coin *j*, the model reads both diaries. Three: it looks at each coin's 10 most recent events, with how long ago they happened turned into a 16-number time code, and an **attention** layer with two heads decides how much each one matters. That gives a 64-number summary per coin. Four: it joins both summaries, both coins' 14 current features and the event's 5 numbers, 161 numbers in all, and a small two-layer network turns them into five chances, one per window. Five: only after a batch of events has been scored does a **GRU** write them into the diaries, and a batch never splits an hour. So the model can never use the future. (After the report we found this made the diaries lag a few hours behind; the "after the report" segment explains the fix.)

It has **70,021 trainable parameters**: about 41,000 in the memory updater, 18,000 in attention and 11,000 in the prediction head. We trained it with the Adam optimizer and a **focal loss**, which focuses learning on the rare yes-cases: up to 100 epochs with early stopping, five times with different seeds, on an ordinary CPU."

IF ASKED

Rebuild 70,021. GRU 41,280: input 149 = 64 + 64 + 5 + 16, three gates × (149 × 64 + 64 × 64 + 64 + 64). Attention 17,984: four 64 × 64 projections with biases plus the event projection 21 × 64. Head 10,693: (161 × 64 + 64) + (64 × 5 + 5). Time encoders 64.

Why sigmoid, not softmax? The five windows are not alternatives: "within 6 hours" includes "within 3 hours". Each window gets its own probability.

Why so small? About 360,000 training events and a noisy target: a bigger network memorises the training years. A smaller and a more regularised version were both worse on validation.

Libraries? PyTorch and PyTorch Geometric (TGNMemory, LastNeighborLoader, TransformerConv), with our own prediction head and training loop.

SEGMENT 8 · 0:50 · HARDIK

Why this design, and the other models

Each competitor answers one question about the main model.

SAY

"We chose the TGN because our data is a stream of time-stamped events between coins: it keeps a memory per coin, reads events in time order and copes with new coins joining. An LSTM only sees one coin's history; a snapshot graph network loses the exact timing.

To test that choice, we raced it against models that each answer one question. Random guessing: what is the floor? An epidemic-style **independent cascade**: does classic network science already do it? **Gradient boosting**: do we need deep learning at all? The TGN **without memory**: does memory matter? **DyGFormer**, a 2023 Transformer: does the result depend on one architecture? For the side tasks we used an LSTM and the classic HAR-RV volatility formula."

IF ASKED

What is the independent cascade? Like a cold going round a class: each shocked coin gets one chance to "infect" each other coin, with chances learned from the training years (Kempe and colleagues, 2003). Test AUPRC 0.159 at one hour.

Sizes? DyGFormer 24,021 parameters; LSTM 5,665.

SEGMENT 9 · 1:15 · HARDIK

Results

The main numbers and five findings, told honestly.

ON SCREEN report Table III (page 5).

Test AUPRC, January 2024 to September 2026 (higher is better)

Model	1 h	3 h	6 h	12 h	24 h
TGN (main)	**0.205**	**0.352**	**0.464**	**0.588**	**0.739**
DyGFormer	0.208	0.351	0.463	0.587	0.736
Gradient boosting	0.178	0.330	0.451	0.581	0.736
Independent cascade	0.159	0.279	0.381	0.507	0.672
Random	0.105	0.206	0.310	0.447	0.627

SAY

"On the test years, at one hour the TGN scores 0.205 against 0.105 for random guessing, about twice as good, and 0.159 for the cascade. Five findings.

One: every model beats random, so there is real signal. Two: the TGN beats the cascade at every window, by 0.046 at one hour and 0.084 at six hours, and in 1,000 bootstrap resamples the cascade never came out ahead. Three: against gradient boosting the TGN is ahead at one and three hours, but only borderline, p about 0.03; boosting had memorised its training years. Four: DyGFormer ties the TGN, so the result is not one architecture's luck. Five: memory helps the slower spread, from 6 to 24 hours, not the next hour.

In practice, the riskiest 1% of alerts came true 32% of the time within an hour and 76% within six hours, against 10% and 31% for random picks. And the graph helps most where contagion is real: after market-wide crashes the TGN scores 0.245 against 0.205 for boosting."

IF ASKED

Isn't 0.205 low? AUPRC is judged against random, not against 1: we are about twice random, and three times random on the top alerts. Most hourly shocks come from news nobody can predict; a much higher score on this task would make us suspect leakage.

Which window matters most? One hour is the headline, because it is the hardest and furthest ahead of random (1.95 times). Six hours is the most useful in practice: time to act, and the biggest lead over the cascade.

SEGMENT 10 · 1:30 · HARDIK

After the report: what we found and fixed

The new work since the report: flaws found, fixes made, and the new best score, said honestly.

SAY

"After submitting, we audited our own pipeline and found real problems. Three mattered. First, the score counted the same coin-hour several times, once per coin that shocked that hour, so busy crash hours dominated. Second, the TGN wrote its memory only once per batch of about 200 events, so 31% of its predictions were made without knowing the previous hour; that is not cheating, the model simply knew less than it could. Third, our inputs were measured in the units of the wild 2020 to 2022 market, so in the calmer test years every coin looked unusual.

We fixed these in a separate copy of the project, made every choice on validation, and scored the test period once. Two results. Our best model is now the TGN with **better inputs**: volatility measured on each coin's own scale, short-window volatility, the hour's price range, trade counts, features from 5-minute bars inside the hour, and the time of day. It scores **0.231** at one hour, up from 0.205, with an AUROC of 0.71. Counting each coin-hour once, it is 2.6 times random. Interestingly, the best version has its memory switched off: the gain came from the inputs, not from memory.

And the biggest finding: when we ask the same contagion question on **15-minute** bars, even a simple model reaches 2.6 times random, and that held on test. Contagion is a minutes-scale process, which is exactly where your network idea should be tested next."

Before and after, test period, 1 hour

Measure	Report (v2 TGN)	Now (v3 TGN)	Random
AUPRC, every event	**0.205**	**0.231**	**0.105**
AUPRC, each coin-hour once (fair)	0.132	0.156	0.059
AUPRC, coins that were calm (no shock in 24 h)	0.052	0.070	0.034
AUROC	0.685	0.709	0.5
AUPRC at 24 hours	0.739	0.762	0.627

IF ASKED

Isn't changing things after seeing results cheating? Every choice was made on validation; each final model was scored on test once, and we list every test score ever taken, old and new, like Table VI.

If memory is off, is it still a graph model? Partly. It still reads each event between two coins, the network features and the market context, but it no longer keeps a diary per coin. Our honest reading: at hourly speed the useful information is the coins' current state and the market's breadth, not their history of interactions.

Has 0.231 been significance-tested? Not yet with the bootstrap; the five seeds agree within ±0.002, and it is ahead of the best simple model (0.218). The bootstrap test is the next step.

Does the dashboard show the new model? No: the deployed dashboard still serves the report's model (seed 0, 0.203). Switching it over is future work.

SEGMENT 11 · 0:55 · DAKSHESH

What drives an alert, and how long shocks last

Explanations and persistence in under a minute.

SAY

"We asked why the model raises an alert using **integrated gradients**, which split a prediction into how much each input pushed it up or down. About 65% comes from the receiving coin's own state, mainly its trading volume compared with its past week and how jumpy it was over the last day; 25% from the coin that shocked; and 10% from the event itself.

We also adapted **TempME**, a 2023 method that looks for chains of past events behind an alert. It found almost none: shocks hit the whole market in the same hour, not along chains, which matches what we saw hour by hour in the FTX collapse.

And shocks are short-lived: a coin's volatility is back to normal after a median of 8 to 10 hours."

SEGMENT 12 · 0:35 · DAKSHESH

The "Secure" part: can it be fooled?

Robustness, attacks and the honest negative result.

> **SAY**
>
> "Losing half of the past events changes the score by under half a percent, and random noise costs little. But a deliberate attack, **FGSM**, that nudges every input by just 0.1 standard deviations in the most harmful direction cuts the one-hour score by about 35%. So a real deployment would need input checks across several exchanges. We also tried a temporal graph autoencoder to flag stressed market days; it did not work, an AUROC of 0.51, and we report that as a negative result."

SEGMENT 13 · 3:45 · DAKSHESH

The dashboard demo, click by click

Five tabs in under four minutes, with what to click, what to point at and what to say.

CAREFUL

- The page opens on **27 September 2026, 23:00 UTC**, the last hour of our data. If the deployed page shows a different hour, read what is on screen; the steps stay the same.
- The opening shock (QNT moving on its own) is the model's weakest situation. Use it to explain the page, then move on.
- Past crises opens on COVID, a training-period crisis with an amber "not a fair test" banner. Switch straight to the bitcoin ETF approval.

1. Opening screen (0:55)

DO point at the left sidebar: *Look ahead* set to 6 hours, the model's run name, "A research tool, not financial advice". Then the top of the page.

> **SAY**
>
> "The page reads the newest prediction from our model. At the top, *What just happened*: at 23:00 UTC on 27 September, QNT dropped about 4.8 times its usual hourly move, on its own, after 9 big moves in the previous day; moves like this usually calm down within about 8 hours.
>
> Below, three numbers: the coin most likely to move next, JUP, with a 37% chance within six hours; a typical coin's chance, 27%; and how many coins are at higher-than-usual risk. The bar chart in *Who's at risk* ranks every coin, coloured against a typical coin, with the dotted line marking typical. '× typical coin' is the chance divided by that typical chance."

2. Why (0:40)

DO open *Why* and pick the riskiest coin.

> **SAY**
>
> "For any coin, the page lists the five inputs that pushed its risk up or down, in plain words with real values, for example trading at well above its usual volume. These come from integrated gradients on the model, so they explain what the model used, not what the market will do."

3. Past crises (1:25)

DO open *Past crises*, choose **Jan 2024 · US spot BTC ETF approval** (green "fair test" banner, test period). Point at "Did the model see it coming?", then press **▶ Play** under the network picture. Then switch to **10 Oct 2025**.

> **SAY**
>
> "Here we replay nine real crises. Each is labelled with its period, because only the test-period ones are true forecasts. For the bitcoin ETF approval in January 2024, at the worst hour, 3 of the model's 5 riskiest coins made a big move within six hours, against 1.6 for five random coins, so the page says 'clearly better than random'. The network replay shows the shock spreading hour by hour.
>
> Now 10 October 2025: here almost every coin moved, so any pick is right, and the page says itself that this crisis cannot judge the model. We built that honesty into the tool."

4. Market map (0:20)

DO open *Market map*.

> **SAY**
>
> "Coins grouped by what they are used for. Lines join coins that moved closely together over the past week; bigger dots have a higher chance; the star marks the coin that just moved."

5. How it works (0:25)

DO open *How it works*, scroll to "How good is it?".

> **SAY**
>
> "Finally, the page is honest about itself: here is the served model's score on the test years at each window against random guessing, and how long different kinds of big moves usually last. That's the dashboard."

IF ASKED

Is it live? No. It shows the newest prediction on our research data, which ends on 27 September 2026, plus replays of nine crises. Updating means running the data pipeline on new Binance hours and then the prediction command; the model itself does not change. Automatic hourly updates are future work.

How is the "chance" made? The model's raw score is calibrated on validation events with an always-rising curve (binned isotonic regression, bins of at least 500), made never to fall as the window grows, and a coin's chance is the highest over the events pointing at it that hour.

Why 0.203 on the page and 0.205 in the report? The dashboard serves one model (seed 0); the report averages five seeds.

Built with? Streamlit and Plotly in Python, reading saved prediction runs; nothing is retrained when the page opens.

SEGMENT 14 · 0:45 · DAKSHESH

Limits and next steps

Say the weaknesses before the professor does.

SAY

"We want to be upfront about the limits. Overall discrimination is modest, an AUROC of 0.64 to 0.69, because most shocks are news-driven. We have one market and one test period, and our coins come from a 2026 ranking, which brings **survivorship bias**.

A simple logistic regression on four numbers matched the report's TGN at one hour, which tells us that at hourly speed most of the signal is in each coin's own state; the new TGN is now ahead of it, 0.231 against 0.218, but that still needs a significance test. And memory, the TGN's signature part, did not help at hourly speed.

Next steps: the TGN on 15-minute bars, where contagion actually happens; a second test period; scheduled-news and order-book data; a money-based backtest; and defences against manipulation."

CAREFUL

Say the logistic-regression result yourself: it costs a little shine but wins trust, and being caught by it costs much more. The answer, if pressed, is question C5.

SEGMENT 15 · 0:25 · HARDIK

Close

One sentence that sums it all up.

> **SAY**
>
> "To sum up: we turned crypto contagion into an honest prediction task, built a temporal graph network that ranks the next coins to be hit more than twice as well as chance (0.231 after our fixes) and beats the classic network model at every window, found that contagion is a minutes-scale process, and explained its alerts in plain words on a dashboard anyone can read. Thank you; we're happy to take questions."

EXPERT

Presenting to the person who gave you the idea

How to talk to an examiner who knows the topic better than you, and the 12 topics he is most likely to probe, each explained in plain words first, then answered.

SIMPLE VERSION

He is not trying to bury you. He gave you a question and wants to see three things: that you understood it, that you tested it honestly, and that you can tell what you found from what you hoped. An expert respects "here is what we expected, here is what the data showed, here is the check we ran" far more than confident claims.

Five rules

1. **Credit the idea, own the work.** "You suggested contagion through networks; these are the choices we made and why."
2. **Scope every claim.** Say "at hourly resolution, on Binance, 2020 to 2026", not "crypto contagion is...". Experts attack overreach, not careful claims.
3. **Surprises are findings.** "We expected Bitcoin to lead; four tests said no." That is research, not failure.
4. **Never bluff.** "We didn't test that. Here is how we would." Then name the test. Reading a number from the report beats guessing it.
5. **Ask back when it is genuine.** "Would you expect lead-lag to show up at minute level?" turns a grilling into a discussion with someone who knows more. Use it once or twice, not as a dodge.

Words to avoid

- "The model predicts prices" (it predicts shocks of either sign).
- "Accuracy" (we use AUPRC).
- "It's live" (it isn't).
- "The graph proves contagion" (it measures and predicts; it does not prove cause).
- "The TGN is much better than everything" (it ties DyGFormer and a simple model).

The 12 topics he will probe

1. Contagion or just moving together?

THE IDEA If all coins fall because the whole market fell, that is **interdependence**: they are always linked. **Contagion** is extra spreading beyond that normal link. Forbes and Rigobon (2002) showed many "contagion" findings were just normal links looking bigger in wild times.

ANSWER "We judge each shock against the coin's own recent volatility and only count shocks that come strictly later. Even so, much of what we see is market-wide news hitting everyone; we say that openly. What we predict is excess risk of a later shock: after a shock, another coin's chance in the next hour rises from 4% to 9.3%, about 2.3 times."

2. Why didn't the spillover network give you directions?

THE IDEA Diebold–Yilmaz fits a **VAR** (each coin predicted from all coins' recent past) and splits each coin's forecast uncertainty into shares caused by each coin. "Sends more than it receives" = transmitter. The **generalized** version (Pesaran–Shin) does not depend on the order of the coins.

ANSWER "At hourly speed BIC picked one lag and the lagged effects are tiny, so the decomposition is driven by same-hour correlation. The NET ranking came out identical to eigenvector centrality (rank correlation 1.00): it tells you how tied a coin is to the market, not who goes first. The daily version (two lags, 10-day horizon) has more room for direction and gives a different ranking (agreement 0.21). Our first draft used the order-dependent Cholesky version, which made Bitcoin look like a giant transmitter; we fixed that."

3. How do you know there is no leader?

THE IDEA Lead-lag means one coin moves first and others follow later. You test it by correlating "Bitcoin now" with "other coin one hour later".

ANSWER "Four checks. Cross-correlation: 0.65 in the same hour, at most 0.035 at any lag, peaking at +24 hours, which is a daily rhythm. A circular-shift null that keeps each coin's habits but breaks timing. Granger tests with Benjamini–Hochberg correction: only 8 of 45 leader-to-coin tests pass, effects about 0.02% of variance, with no consistent direction. And replication: the training pattern does not repeat on validation. Our rule required all four, so we added no directed edges."

4. Why a threshold of 0.75 for the correlation network?

THE IDEA Draw a line between coins whose correlation is above a cut-off. Too low and everything is joined to everything.

ANSWER "With week-long windows almost every pair clears 0.5, which gave a 75%-dense graph with no structure. We fitted the cut-off on training data to match the daily graph's 30% density: 0.75 at one hour, 0.77 at two. Each coin also keeps its two strongest links. The Epps effect, correlations looking weaker over short intervals, is weak in crypto (0.58 at one hour, 0.59 at a day), so one rule works at both speeds."

5. Did the network features actually help the model?

THE IDEA We fed 8 network numbers per coin (centrality, spillover, shock PageRank and so on) into the TGN alongside 6 market numbers.

ANSWER, HONESTLY "Only a little at one hour. Attribution says most of an alert comes from the coin's own recent trading, and the TGN without memory is just as good at one hour. Where the network adds is slower spread (memory helps at 6 to 24 hours) and broad crashes (0.245 against 0.205 for boosting after market-wide crashes). Network rankings also drift over the years, which makes them weaker as fixed inputs."

6. If contagion is simultaneous, why a graph model at all?

THE IDEA A fair challenge: if everyone moves in the same hour, maybe the graph is not needed.

ANSWER "At one hour, a simple model ties the TGN; we say that. But the graph model beats the classic network-diffusion model at every window, its memory helps from 6 to 24 hours, and it does best exactly where contagion is broad. Our honest conclusion is that hourly data hides most of the ordering; finer data, minutes or order books, is where a graph model should pull ahead."

7. How was the independent cascade fitted?

THE IDEA An epidemic model: each shocked coin gets one chance to "infect" each other coin, with a probability per pair.

ANSWER "For each pair, how often coin j shocked within k hours after coin i, on training data only, pulled towards j's normal rate as if we had 10 extra observations, so rare pairs don't get extreme numbers. With several coins shocked: 1 minus the product of the escape chances. It scores 0.159 at one hour; its pair numbers alone do no better than the base rate, so its gain comes from counting how many coins shocked at once. That told us the market-wide signal matters more than pairs."

8. What does the market mode mean?

THE IDEA Find the main directions in which 30 coins move; one is "everything together". Random-matrix theory (Marchenko–Pastur) says how big an eigenvalue pure noise could produce.

ANSWER "The biggest eigenvalue is 19.3 of 30, so 64% of hourly movement is the market as a whole; it is the only one above the noise bound of 1.08. That is why we removed it before finding communities, otherwise everything is one group."

9. Do communities match sectors?

THE IDEA Louvain finds groups of coins more tied to each other than to the rest; NMI measures agreement with sector labels (0 to 1).

ANSWER "Partly: NMI 0.47 against 0.32 for shuffled labels, $p = 0.003$. Meme coins form their own group; old payment coins another. But after a shock, contagion happens equally inside and across sectors (9.3% and 9.3%), except among meme coins (13.7%)."

10. Why the minimum spanning tree, and what did it show?

THE IDEA The MST keeps the 29 strongest links joining all 30 coins with no loops; its average length shrinks when coins move as one (Mantegna, Onnela).

ANSWER "It shrank in every crisis, to its lowest, 0.36, in April 2024, so diversification vanishes exactly when needed. DOT has the most links, but to loosely attached coins, so it is a meeting point for outsiders, not a hub of influence."

11. Why not a normal graph neural network on snapshots?

THE IDEA A snapshot GNN looks at one picture of the network per hour or day.

ANSWER "Contagion is about order in time: who shocked, when, and what happened since. A snapshot loses that order and every coin's history between pictures. The TGN reads events one by one in time with a memory per coin. We did use snapshot graphs, for the T-GAE anomaly model, and it did not detect stress days."

12. How would you take the idea further?

THE IDEA Show you can see beyond what you built.

ANSWER "Minute-level data, to test whether a leader appears inside the hour; directed networks built from that ordering; a second test period; scheduled-news timing; and a money-based test of the alerts. Your network idea is strongest where time is fine enough to see who moves first, and hourly bars blur that."

REFERENCE

Every model we used, in one place

What each model is, why it is there, and its score. Test AUPRC at 1 hour unless marked; random is 0.105.

Model	What it is, simply	Why it is there	Score
Contagion: will coin j shock within k hours after coin i?			
Random / base rate	each coin's usual chance	the floor	0.105
Independent cascade	epidemic model: each shocked coin gets one chance to "infect" each other coin, chances learned from training years	classic network-science baseline	0.159
Gradient boosting	hundreds of small decision trees, each fixing the last; same inputs as the TGN's head	do we need deep learning?	0.178
Boosting, retrained on train + validation	the same, more data	does recent data help?	0.185
Ensemble of TGN, DyGFormer, boosting	ranks averaged with weights chosen on validation	does combining help?	0.199
TGN v2 (the report's main model)	memory per coin (GRU), attention over 10 recent events, head 161 → 64 → 5; 70,021 parameters	main model	0.205
TGN without memory (v2)	same with memory and attention off	does memory matter?	0.208
TGN retrained on train + validation	fixed 77 epochs, no early stopping	does recent data help?	0.188
TGN on 2-hour bars	same model, coarser data	is 1 hour the right speed?	0.165*

Model	What it is, simply	Why it is there	Score
DyGFormer	a small Transformer that rereads each coin's last 16 events (NeurIPS 2023); 24,021 parameters	does the result depend on one design?	0.208
Logistic regression, 4 numbers	one weighted sum: volume surge, hours since last shock, market breadth, size of the move	found in our review: a very simple rival	0.213
Boosting, new inputs + 5-minute features	after-report version	do better inputs help trees?	0.210
Logistic regression, new inputs + 5-minute features	after-report version	do better inputs help a simple model?	0.218
TGN v3 (best now)	**TGN with 20 better inputs, shock sizes in messages, time of day and market context; memory off; 5 seeds**	**after-report best model**	**0.231**
Logistic regression / boosting on 15-minute bars	the same question with 15-minute shocks and windows	is contagion faster than an hour?	0.311 / 0.282†
Side tasks and tools			
LSTM	reads one coin's last 168 hours; 5,665 parameters	next-hour shock of one coin (T2)	0.109‡
HAR-RV	straight-line formula from last hour's, day's and week's volatility	volatility forecast (T3); beat the LSTM	–

Model	What it is, simply	Why it is there	Score
Diebold–Yilmaz VAR	splits each coin's forecast uncertainty into shares from each coin	spillover network, NET ranking	–
T-GAE	graph autoencoder that predicts tomorrow's network; 3,646 parameters	flag stress days: failed (AUROC 0.51)	–
Integrated gradients, TempME	explainers: which inputs, which past events drive an alert	why the model alerts	–

* first 2-hour bar. † 15-minute window, random 0.120: not comparable with hourly scores; compare lift (2.6×). ‡ different task, random 0.039.

REFERENCE

Every notebook in two minutes

If the professor says "show me notebook 02", know its question, its three findings and the chart to open. (Guide Part 10 has every section.)

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01 · The data and a first look

QUESTION Is the data clean, and how does crypto behave?

FINDINGS Prices wander, returns don't (ADF test), so models use returns. Returns have fat tails (kurtosis 18) and no direction memory, but jumpiness clusters, so we predict shocks, not direction. Rush hours (Wednesday 12:00 UTC is 2.17× an average hour) are removed. No coin leads at hourly speed (lagged correlation 0.035). After a shock, another coin is 2.3× more likely to shock in the next hour.

SHOW the contagion-jump bar chart (section 17) and the lead-lag chart (section 15).

02 · The network of coins

QUESTION Who is connected, who is central, how do crises change the network?

FINDINGS One market factor explains 64%. Hourly spillover ranking equals plain centrality. Communities partly match sectors (NMI 0.47), contagion ignores sectors except meme coins. The tree shrinks in every crisis. FTX: 18 to 24 coins shocked in the same hour as FTT. The cascade baseline scores 0.159.

SHOW the MST picture (section 5) and the FTX hour-by-hour chart (section 8).

03 · The model race

QUESTION Does the TGN beat simpler methods, honestly?

FINDINGS Beats the cascade at every window (bootstrap). Ties DyGFormer. Borderline over boosting at 1 to 3 hours. Memory helps only from 6 hours. Top 1% alerts: 32% at 1 hour, 76% at 6. Retraining on more data hurt the TGN. HAR-RV beats the LSTM on volatility.

SHOW the main results table (section 1) and the bootstrap tables (section 4).

04 · Why does the TGN alert?

QUESTION Which inputs and which past events drive an alert?

FINDINGS Receiving coin 65%, shocking coin 25%, event 10%. TempME barely beats random (0.157 against 0.152); the TGN's attention is perfectly even; event patterns look like a shuffled stream. 13 of the 15 top alerts in three crashes came true.

SHOW the attribution bar chart (section 5).

05 · How long does a shock last?

QUESTION How long, and does the type of shock matter?

FINDINGS Back to normal in 8 to 10 hours; extra jumpiness halves in 2.5 hours. 55 to 65% re-shock within a day. Market-wide crashes pull in about 8 coins within 6 hours, isolated shocks about 5. The TGN beats boosting after broad shocks, not after isolated ones.

SHOW the |z| profile chart (section 2).

06 · Can it be fooled?

QUESTION Robustness, and can an autoencoder spot stress days?

FINDINGS 50% missing events: under 0.5% change. Noise of 0.5: −5.7%. FGSM attack of 0.1: −35%. The T-GAE failed (0.51 against 0.89 for turbulence) but ranked 10 October 2025 as the strangest day.

SHOW the perturbation chart (section 2).

archive/ · Our first drafts

WHY IT'S THERE Kept with a list of every error: a shock measured partly against itself, same-day co-occurrence counted as contagion, an order-dependent spillover method. All fixed; it shows the corrections were honest.

After the report · the v3 scripts (sandbox branch)

WHAT THEY DO `fair_report.py` rescores models counting each coin-hour once; `fetch_5m.py` downloads 19.3 million checksum-verified 5-minute bars; `go_nogo_5m.py` tests the new inputs on the hourly task; `e2_15min.py` asks the contagion question at 15 minutes; `e2_sender_check.py` tests whether knowing which coin shocked helps. Write-up: `docs/accuracy_v3_notes.md`.

REFERENCE

All the findings, in one list

Everything the project found, report and after, one line each.

About the market

1. One market factor explains 64% of hourly movement.
2. Coins move together within the hour; no coin leads (lagged correlation 0.035).
3. After a shock, another coin is 2.3× more likely to shock in the next hour; the effect fades within a day.
4. In every crisis, correlation rises and the network tightens; diversification disappears.
5. Contagion ignores sectors, except among meme coins.
6. Shocks fade in 8 to 10 hours, and cluster: 55 to 65% re-shock within a day.
7. Contagion is a minutes-scale process: at 15-minute bars even simple models reach 2.6× random.
8. Knowing which coin shocked adds only a little beyond how many shocked (+0.013 at 15 minutes), because the coins are so correlated.

About the models

1. The report's TGN scores 0.205 at 1 hour (2× random) and beats the network-diffusion baseline at every window.
2. It ties DyGFormer and a simple 4-number logistic regression at 1 hour; it is only borderline over boosting.
3. Memory helped only from 6 hours in the report, and not at all after the fixes.
4. Alerts are driven by the receiving coin's own state; there are no special chains of events.
5. The model shrugs off missing data and noise but is fooled by deliberate attacks (−35%).
6. With better inputs, the TGN (memory off) reaches 0.231 on test, 2.6× random when each coin-hour counts once.
7. Validation gains do not always survive the test: boosting's did not; the new TGN's did.

REFERENCE

What we would do next

Improvements in order of how much they should matter. Say the first three if asked "what next?".

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1. **Run the TGN on 15-minute bars.** Contagion happens within minutes, and that is where the graph model and the professor's network idea should finally beat simple models.
2. **Test the new model properly.** A day-block bootstrap of v3 against the simple model, and report the fair score (each coin-hour once) as the standard.
3. **Train longer.** The final v3 runs picked epochs 57 to 59 of 60, so they were still improving.
4. **A second test period.** Rolling test windows, retraining every few months as the market changes; six coins had no training data at all.
5. **New information.** A calendar of scheduled US news, order books and trade-by-trade data, and news or social-media signals.
6. **Make memory earn its place,** or drop it: other ways to combine an hour's messages, or a design built for simultaneous events.
7. **Prove it is useful.** Check calibration on test, and a money-based backtest: does acting on alerts reduce losses?
8. **Defend it.** Adversarial training and checking inputs across several exchanges.
9. **Ship it.** Serve v3 on the dashboard and turn on the live scanner we built, which fetches new Binance hours and logs every alert.
10. **Fix survivorship bias.** Include coins that were delisted or collapsed.

RECAP

If the professor opens the report

The report in one page: structure, research questions, the equations to know, and the small mismatches to be ready for. (Guide Part 12 has every detail.)

Structure (9 pages, IEEE style, 19 October 2026, 38 references): I Introduction (problem, why deep learning, four research questions, four contributions) · II Literature review (contagion and spillovers, network science, temporal graph learning, explainability and evaluation, the gap) · III Methodology (data and Table I, shock and label equations 1 to 3, network analysis, TGN equations 4 to 7 with Figures 1 and 2, training with equation 8 and Table II, scoring with equations 9 and 10) · IV Results (tuning in Figure 3, test results in Table III and Figures 4 and 5, Table IV, shock types in Table V and Figure 6, explanations, robustness in Figure 7, limitations, Table VI) · V Conclusion · Acknowledgment (AI use, reproducibility).

Research questions and answers

Question	Answer
RQ1. Do temporal graph models beat a fitted network-diffusion model?	Yes, at every window (+0.046 at 1 h, +0.084 at 6 h).
RQ2. Do they beat plain machine learning, and does memory matter?	Only borderline against boosting at 1 to 3 h; memory helps from 6 h.
RQ3. What drives their alerts?	The receiving coin's current state (about 65%).
RQ4. How long do shocks last; how robust is the model?	8 to 10 hours; robust to gaps and noise, not to attacks (−35%).

The equations most worth practising

#	In plain words
1–2	Shock score z = adjusted return ÷ typical move over the previous 168 hours; shock if |z| > 2.29.
3	Label = 1 if coin j shocks in hours t + 1 to t + k; never the same hour.
4	Message = i's memory (64) ‖ j's memory (64) ‖ the event's 5 numbers ‖ time code (16) = 149 numbers.
5	New memory = GRU(old memory, message).
6–7	Attention over 10 recent events gives 64 numbers per coin; the head turns 161 numbers into 5 chances (sigmoid).
8	Focal loss: easy cases count less; γ 2, α 0.25.
9–10	AUPRC as average precision; the bootstrap difference between two models over resampled days.

Table VI lists every version ever scored on the test period, including the ones we did not use (TGN v1 0.201, three no-memory versions, the refits): it shows we did not keep trying until something scored well.

SMALL MISMATCHES TO KNOW

- Table I counts 1,463,693 events; the models read 1,463,147, because the 546 shocks in the first week of 2020 lack a week of history.
- Section III-C says eigenvector centrality was computed on the directed networks; it was computed on the correlation network (PageRank used the directed ones).
- The report's 197 tests passed and 5 skipped were counted at submission; the deployed branch has 210.
- Dakshesh's PRN is blank on the cover: fill it in by hand on a printed copy.

Q&A

Questions you will get

54 likely questions with answers short enough to say out loud. Turn on quiz mode, answer first, then reveal.

A. Problem, data and events

A1 How did you quantify events?

In four steps. A shock is an hour where a coin's rush-hour-adjusted return is more than 2.29 times its typical hourly move over the previous 168 hours. Each shock of coin i creates one event for every other active coin j. Each event carries 5 numbers (two correlations between i and j, i's return, i's volatility, a shock flag), and each coin has 14 features. Each event gets 5 labels: did j shock within 1, 3, 6, 12, 24 hours, strictly later. 1.46 million events in total.

A2 Why 2.29, and why 168 hours?

2.29 was set on training data so that about 4% of hours are shocks: rare, but frequent enough to learn from. 168 hours is one week, which covers every weekday and the weekend and is stable without being slow to adapt. The window ends the hour before, so a move is never judged against itself.

A3 Why must the other coin's shock come strictly later?

Coins moving in the same hour is co-movement, not contagion. Forbes and Rigobon showed that confusing the two makes contagion look bigger than it is. So the label counts hours t + 1 to t + k only.

A4 Why hourly data?

Contagion happens within hours: after a shock, another coin is 2.3 times more likely than usual to shock in the very next hour, and the effect fades within a day. Daily data merge several shocks and their order.

A5 How did you avoid data leakage?

Fixed date splits with a 7-day embargo; every scaler, threshold and seasonal profile fitted on training data only; shocks measured against the past only; labels strictly later; the model's memory updated only after an hour is scored; tuning on validation; the test period scored once.

A6 Why these 30 coins?

The largest coins by market size (CoinGecko, 27 September 2026), excluding stablecoins and wrapped tokens, that trade against USDT on Binance with at least two years of hourly history and 90% coverage of the test period. LUNC and FTT are kept only for crash case studies. The weakness is survivorship bias, which we admit.

A7 What are the 14 coin features?

Six market features: last hour's return, volatility over 24 and 72 hours, trading volume, volume compared with its own past week, and how long the coin has been listed. Eight network features: strength, eigenvector centrality, minimum-spanning-tree links, community size, spillover PageRank and NET, shock PageRank and shock out-lift, all from past data only.

A8 What data problems did you hit?

MATIC renamed to POL (a 1:1 swap, gap masked); the LUNA ticker reused for a new coin after the collapse (never stitched together); FTT delisted after FTX; maintenance gaps (up to 2 hours filled and flagged, longer masked); timestamps switching units in 2025; four days when USDT slipped from $1 (flagged).

A9 Why do intraday patterns matter?

Markets have rush hours: Wednesday 12:00 UTC is about twice as jumpy as an average hour, Sunday morning much calmer. We divide that pattern out before measuring shocks, otherwise every busy Wednesday afternoon would look like a shock.

A10 Why can't you predict whether prices go up or down?

Direction has almost no memory (today's sign matches yesterday's 47% of the time), but jumpiness does: calm follows calm and storms follow storms. So we predict shocks of either sign, which is predictable, rather than direction, which is not.

B. The model and training

B1 What is a TGN, who made it and when?

A Temporal Graph Network, by Rossi, Chamberlain, Frasca, Eynard, Monti and Bronstein at Twitter with Imperial College London, published in June 2020. It keeps a memory per node, updated by a GRU at each event, and reads recent neighbours with time-aware attention.

B2 Why did you choose the TGN?

Our data is a stream of time-stamped events between coins. The TGN keeps a memory per coin, reads events in time order and handles coins joining later. An LSTM sees one coin at a time; a snapshot graph network loses the exact timing; trees see one fixed snapshot of features.

B3 How many parameters, and where are they?

70,021: the GRU memory updater 41,280, attention 17,984, the prediction head 10,693, time encoders 64.

B4 Explain memory, message and attention in simple words.

Memory is a 64-number diary per coin. A message is what gets written into the diary after an event: both diaries, the event's 5 numbers and how long since the last entry, 149 numbers. Attention decides how much each of a coin's 10 most recent events should count when summarising the coin right now.

B5 How does the model avoid seeing the future?

Events are processed in time order in batches of about 200 that never split an hour; a batch is scored before its events are written into memory, so no event sees its own hour. The catch, found after the report: a batch can span several hours, so memory sometimes lagged (question F3).

B6 What is focal loss, and why use it?

A loss that gives less weight to easy, obvious examples and more to rare, hard ones ($\gamma = 2$, with weight 0.25 on the yes-cases). Only about 10% of events are yes at one hour, so ordinary loss would be dominated by easy no-cases.

B7 How was it trained?

Adam, learning rate 0.0005, weight decay 0.0001, batches of about 200 events, up to 100 epochs, stopping after 10 epochs without improvement in validation AUPRC at one hour, keeping the best epoch. Five seeds, on a CPU, about half a minute per epoch.

B8 Why sigmoid outputs rather than softmax?

The five windows are not competing answers: a shock within 3 hours is also within 6. Each window gets its own probability.

B9 What is DyGFormer and why include it?

A 2023 Transformer model for event streams (Yu and colleagues, NeurIPS) with no memory: it rereads each coin's last 16 interactions every time. We rebuilt a compact version (24,021 parameters) on exactly our events. It ties the TGN, so our result does not depend on one architecture.

B10 Did you try to improve the TGN?

Yes, on validation only. Training longer helped most (0.138 to 0.153). A learning-rate schedule, extra context features, stronger regularisation, a smaller network and futures-market data did not beat that, so we kept the simplest well-trained version.

C. Results and honesty

C1 Isn't an AUPRC of 0.205 low for deep learning?

AUPRC is judged against random guessing, which scores the yes-rate, 0.105 here: we are about twice random, and our top 1% of alerts come true 32% of the time against 10%. Most hourly shocks are triggered by news, so no model can see them coming; a much higher score on this task would make us suspect leakage.

C2 Why not accuracy?

Always saying no is 89.5% accurate at one hour and catches nothing. AUPRC rewards ranking the real yes-cases at the top.

C3 How do you know the TGN beats the cascade by more than luck?

A day-block bootstrap: we rebuilt the test period 1,000 times from whole days and rescored both models on the same events. The 95% interval at one hour is 0.019 to 0.074, above zero, and the cascade never won a single rebuild.

C4 Does memory matter?

Not for the next hour (no difference), but yes for slower spread: +0.010 to +0.013 from 6 to 24 hours, significant.

C5 Did you compare with a simple model?

Yes. The cascade and boosting are in the report, and in our final review a logistic regression on four numbers (the coin's volume against its past week, hours since its last shock, how much of the market shocked in the last day, and the size of its current move) matched the TGN on test. It confirms what our explanations show: at hourly speed the signal is mostly in each coin's own state, because shocks hit the market in the same hour. The TGN learns this from the raw event stream and clearly beats the network model; finer data such as minute bars is where a graph model could pull ahead.

C6 Is the TGN overfitting?

Somewhat at short windows: its edge over random shrinks more from training to validation than the cascade's (a ratio of 1.5 against 1.1), but not at 24 hours. Early stopping keeps it in check; more regularisation only lowered validation scores.

C7 Why is test better than validation (0.205 against 0.153)?

The test years had more shocks and more moving together: the yes-rate is 10.5% against 8.6%, and even the cascade scores higher on test. Against random, the gap is small (1.95 times against 1.78).

C8 What drives an alert?

Integrated gradients: about 65% the receiving coin's own state (volume surge against its past week, 25%; 24-hour volatility, 12%; latest return, 9%), 25% the shocking coin and 10% the event. The model flags coins that are already stretched when another coin moves.

C9 What did TempME find?

Almost nothing: its explanations were barely better than random picks (0.157 against 0.152), because the model's attention is spread evenly and event patterns look the same as in a time-shuffled stream. Shocks spread across the market in the same hour, not along chains. That is a finding about the market, not a bug.

C10 How robust is it?

Missing up to half the events: under 0.5% change. Random noise of half a standard deviation: −5.7%. A deliberate FGSM attack of 0.1 standard deviations: about −35%. The risk is manipulation, not ordinary data errors.

D. The dashboard

D1 Is the dashboard live?

No. It shows the latest prediction on our research data, which ends on 27 September 2026, plus replays of nine crises. Updating it means running the pipeline on new Binance hours; the model does not change. Automatic hourly updates are future work.

D2 How is a coin's "chance" computed?

The TGN's raw score is calibrated on validation data with an always-rising curve (binned isotonic regression), forced not to fall as the window grows; a coin's chance is the highest over that hour's events pointing at it.

D3 What does "× typical coin" mean?

The coin's chance divided by a typical coin's chance after a shock: 1.4 × means 40% more likely than usual.

D4 The model beat random clearly in only one crisis. Why?

In big crashes almost every coin moves, so any five picks are right and nothing can be judged; the page says so. Six of the nine crises were in the training or validation years, so they are not fair tests. Of the three test-period crises, the bitcoin ETF approval is the one where picks could be told apart, and there the model got 3 of 5 against 1.6 for random.

D5 Why does the page show 0.203 and the report 0.205?

The dashboard serves one trained model (seed 0); the report averages five seeds.

D6 How was the dashboard built?

Streamlit and Plotly in Python. A prediction command replays the event stream through the trained TGN, saves chances and reasons per coin, and the page reads those files; nothing is retrained when it opens. Reasons are integrated gradients turned into plain words with the coin's real values.

E. The report and the project

E1 What are your research questions and answers?

RQ1, beat a network-diffusion model? Yes, at every window. RQ2, beat plain machine learning, and does memory matter? Borderline at 1 to 3 hours; memory helps from 6 hours. RQ3, what drives alerts? The receiving coin's state. RQ4, how long do shocks last and how robust is it? About half a day; robust to noise, not to attacks.

E2 What is new in your work?

Finance studies measure how connected coins are; graph-learning studies test models on "will two users interact?". We built a fair, hourly, no-peeking benchmark for coin-to-coin contagion, raced temporal graph models against finance, network and machine-learning baselines, and analysed what the models learn.

E3 Explain equation 4.

The message that updates coin i's memory: i's memory just before the event, j's memory, the event's 5 numbers and a 16-number code of the time since i's last update, glued together: 149 numbers. Equation 5 feeds it to a GRU, which writes the new memory.

E4 What is Table VI?

Every version we ever scored on the test period, including superseded and rejected ones. It shows we did not keep trying versions until one scored well; the main model was fixed from validation evidence.

E5 What are the main limitations?

Modest discrimination (AUROC 0.64 to 0.69); one market and one test period; survivorship bias in the coin list; calibration not checked on the test period; no money-based evaluation; no news, social-media or order-book data.

E6 How would you improve the one-hour score?

Give the model the "already hot" signals directly, score each coin once per hour, and blend it with a simple model: in a check after the report this raised validation AUPRC from 0.153 to about 0.168. Bigger gains need new information: minute-level timing inside the hour and a calendar of scheduled news.

E7 Who did what?

Agree this before the viva. Suggested: Hardik built the data pipeline, the network analysis, the models and evaluation, the explanations and the robustness work; Dakshesh built the prediction command, the dashboard and its crisis explorer, the model compression, the extra attack tests and part of the persistence analysis.

E8 Did you use AI tools?

Yes, as the report's acknowledgment says: an AI assistant helped with code, cross-checking numbers against the run logs, re-running tests and drafting the report. We set the questions, made the choices, checked the results, and can explain every part.

F. The work after the report

F1 What changed after the report?

We audited our own pipeline, fixed the memory timing and message handling, rebuilt the inputs on each coin's own scale with short-window, price-range, trade and 5-minute features, and added a fair score that counts each coin-hour once. The new TGN scores 0.231 on test at 1 hour, up from 0.205.

F2 Isn't improving after seeing the test result cheating?

Every choice was made on validation, including switching memory off (0.179 against 0.168 on validation). Each final model was scored on test once, and we list every test score ever taken. Some ideas failed on test, such as boosting with the 5-minute features, and we report that too.

F3 What was wrong with the memory?

It was written once per batch of about 200 events, which can span several hours, so 31% of predictions lacked the previous hour. And because an hour's events share one timestamp, only one arbitrary message per coin was kept. It never used future data; it just knew less than it could. We now write memory every hour and average the hour's messages.

F4 Your best model has memory switched off. Is it still a graph model?

Partly: it still scores each event between two coins, with network features and the market's breadth, but it keeps no diary per coin. That is a finding: at hourly speed, what matters is the coins' current state and how many coins are shocking, not their history of interactions.

F5 What are the new inputs?

20 per coin: volatility over 24, 72 and 168 hours and over 3 and 6 hours, the same measured against the coin's own weekly level (so calm years don't look odd), the hour's price range, trade counts against the past week, and four features from 5-minute bars (5-minute volatility, how much of the move was one jump, and the last 15 minutes' move). Plus time of day, weekend, shock sizes in the messages and the market context.

F6 What is the "fair" score?

The label depends only on the receiving coin and the hour, so one coin-hour appeared about 3.7 times on test, once per shocking coin. Counting each once: v2 0.132 and v3 0.156 against 0.059 random, 2.2× and 2.6×.

F7 What is the 15-minute result?

The same contagion question on 15-minute bars, with shocks defined the same way. A simple logistic regression reaches 0.311 against 0.120 random at 15 minutes (2.6×, AUROC 0.74), and validation agreed (0.29). It uses the receiving coin's state and market-wide numbers only, so it shows contagion is fast; it is not yet a graph result.

F8 Does knowing which coin shocked help?

A little. At 15 minutes, adding the receiver's correlation and lead-lag with the coins that shocked raised boosting by 0.013 (95% interval 0.002 to 0.021) on validation. Small, because these coins are so correlated that who shocked says little beyond how many shocked.

F9 Why did the new inputs help the TGN on test but not boosting?

We don't know for sure. Boosting fitted its validation gain too closely; the neural model with five seeds kept its gain (0.179 on validation, 0.231 on test, seeds within ±0.002). A bootstrap test against the simple model is the next step before calling it a clear win.

F10 Is the new work in the dashboard and the code?

Not yet. The deployed dashboard serves the report's model. The new code is on a separate branch with tests (37 passing, and the old settings reproduce the original run exactly); it will be committed and the dashboard switched over after review.

NUMBERS

Numbers card

The figures to have on the tip of your tongue.

Topic	Number
Data	30 coins · Binance hourly · 1 Jan 2020 to 27 Sep 2026 · 59,088 hours · 3,049 files checked
Splits	train to Jun 2022 · validation Jul 2022 to Dec 2023 · test Jan 2024 to Sep 2026 · 7-day embargo
Shock	|z| > 2.29 against the previous 168 hours · about 4% of hours
Events	1.46 million · 5 numbers per event · 14 features per coin · 5 labels
Yes-rate (test)	10.5% · 20.6% · 31.0% · 44.7% · 62.7%
TGN	Rossi et al., Twitter, 2020 · 70,021 parameters · memory 64 · time code 16 · 10 recent events · 2 heads · head 161 → 64 → 5
Training	Adam, learning rate 0.0005 · focal γ 2, α 0.25 · up to 100 epochs, patience 10 · 5 seeds · CPU
TGN test AUPRC	**0.205 · 0.352 · 0.464 · 0.588 · 0.739 (random 0.105 · 0.206 · 0.310 · 0.447 · 0.627)**
Against rivals at 1 h	cascade 0.159 · boosting 0.178 · DyGFormer 0.208 · no memory 0.208
Bootstrap	vs cascade +0.046 (0.019 to 0.074) at 1 h; vs boosting p about 0.03 (borderline)
Top 1% alerts	32% at 1 h · 61% at 3 h · 76% at 6 h (random 10.5% · 21% · 31%)
After market-wide crashes, 1 h	TGN 0.245 vs boosting 0.205
Explanations	receiver 65% · source 25% · event 10%
Persistence	back to normal in 8 to 10 h · 55 to 65% re-shock within a day
Robustness	50% missing events: under 0.5% · FGSM 0.1: −35% · T-GAE AUROC 0.51
Dashboard	5 tabs · serves the report's model, seed 0 (0.203 at 1 h) · bitcoin ETF: 3 of 5 vs 1.6 random at 6 h

Topic	Number
After the report: v3 TGN	**0.231 · 0.382 · 0.499 · 0.619 · 0.762 · AUROC 0.709 · memory off · 5 seeds within ±0.002**
Fair score (each coin-hour once), 1 h	v2 0.132 · v3 0.156 · random 0.059 (2.2× → 2.6×)
Simple rivals, test 1 h	logistic 0.213 (old inputs), 0.218 (new) · boosting with 5-minute features 0.210
Memory flaw	written per ~200-event batch; 31% of predictions missed the previous hour
15-minute task	0.311 vs 0.120 random (2.6×), AUROC 0.74 · validation 0.289 · sender information +0.013

BACKUP

If something goes wrong

Plan B for the room.

- **The dashboard will not load or is asleep:** "It's waking up; meanwhile, here are screenshots of each tab." Carry on with the screenshots and return to the page when it loads.
- **A question you cannot answer:** "We didn't test that. Here is how we would: ..." Then name the experiment. Never guess a number.
- **You forget a number:** open the report at Table III or the numbers card; reading it is better than inventing it.
- **"This is not real time":** agree, and give the "is it live?" answer from segment 13 (the demo).
- **"A simple model does as well":** agree, explain it (question C5), and point to what the TGN adds: it learns this from the raw event stream, it beats the network-diffusion model at every window, and it does best after market-wide crashes.
- **Running out of time:** jump to segment 15 and close; the questions will bring out the rest.
- **The two of you disagree on an answer:** let the person who built that part answer, and support them; never correct each other in front of the examiner.

CryptoGraph Presentation Script · prepared on 28 October 2026 for the deployed dashboard (GitHub branch `feat/hourly-network-models`) and the CA3 report of 19 October 2026. Local only; keep it out of the repository's commits, like the viva guide.